INCH-POUND

MIL-C-24643A
AMENDMENT 1
22 November 1994

MILITARY SPECIFICATION

CABLE AND CORDS, ELECTRIC LOW SMOKE, FOR SHIPBOARD
USE GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-C-24643A, 14 March 1994, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

1.2.1 Delete and substitute:

"1.2.1 Watertight, (with circuit integrity), nonflexing service. Cable and cord types for watertight, (with circuit integrity), nonflexing service as follows:

Electronic, communication and instrumentation:

	Unarmored	Unarmored with overall shield	Armored		
Type	LSTTSU	-----	LSTTSA	1-1/2 through 60 pairs	300V"

PAGE 3

1.2.3 Delete and substitute:

"1.2.3 Non-Watertight, nonflexing service. Cable and cord types for non-watertight, nonflexing service are as follows:

Power & Lighting:

	Unarmored	Unarmored with overall shield	Armored		
Type	----	-----	LSTNWA 2/	Two conductor with overall shield	600V"
	----	LS2SJ	LS2SJA		

Control:

	Unarmored	Unarmored with overall shield	Armored		
Type	LSMU	LSMUS*	LSMA	14 conductors	300V"

Electronic, communication and instrumentation:

	Unarmored	Unarmored with overall shield	Armored		
Type	----	LS3SJ	LS3SJA	Three conductors with overall shield	300V"

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1.2.4 Delete and substitute:

"1.2.4 Non-Watertight, flexing service. Cable and cord types for non-watertight, flexing service are as follows:

Power and lighting:

Type	Unarmored	Unarmored with overall shield	Armored		
	LSSRW	-----	LSSRWA	Single conductor	300V"

TABLE III - Delete and substitute:

"TABLE III. Diameters, area and weights of concentric lay stranded copper conductors, ASTM B 8

Size of conductor, circular mils or AWG numbers	Nominal cond dia. class B (inches)	Area (square in.)	Approx. weight lbs per 1000ft	Approx. weight lbs per mile
2,000,000 cmil	1.632	1.571	6,175	32,600
1,600,000 cmil	1.459	1.257	4,940	26,080
* 1,300,000 cmil	1.315	1.021	4,014	21,190
1,000,000 cmil	1.152	0.7854	3,088	16,300
800,000 cmil	1.031	.6283	2,470	13,040
650,000 cmil	.929	.5105	2,007	10,600
550,000 cmil	.855	.4320	1,698	8,966
350,000 cmil	.681	.2749	1,081	5,706
300,000 cmil	.630	.2356	926.3	4,891
250,000 cmil	.575	.1963	771.9	4,076
No. 0000	.528	.1662	653.3	3,450
No. 000	.470	.1318	518.1	2,736
No. 00	.419	.1045	410.9	2,169
No. 0	.373	.08289	325.8	1,720
No. 1	.332	.06573	258.4	1,364
No. 2	.292	.05213	204.9	1,082
No. 3	.260	.04134	162.5	858.1
No. 4	.232	.03278	128.9	680.5
No. 5	.206	.02600	102.2	539.6
No. 6	.184	.02062	81.05	428.0
No. 7	.164	.01635	64.28	339.4
No. 9	.130	.01028	40.42	213.4
No. 10	.116	.008155	32.06	169.3
No. 12	.0915	.005129	20.16	106.5
No. 14	.0726	.003225	12.68	66.95
No. 16	.0576	.002028	7.974	42.10
No. 18	.0456	.001276	5.015	26.48
No. 20	.0363	.0008023	3.154	16.65

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TABLE IV: Delete and substitute:

"TABLE IV. Construction requirements of rope lay stranded copper conductors having bunch stranded members[1/] ASTM B 172.

		Class I		Class K		Class M	
Area of cross section (circular mils)	Size (AWG)	Nom. no. of strands 0.0201 inch in dia. (no.24 AWG)	Strand Const.	Nom. no. of strands 0.0100 inch in dia. (no. 30 AWG)	Strand Const.	Nom. no. of Strands 0.0063 inch in dia. (no. 34 AWG)	Strand Const.
650,000	---	1,596	19x7x12	6,517	19x7x49	16,226	61x7x38
600,000	---	1,470	7x7x30	5,985	19x7x45	14,945	61x7x35
500,000	---	1,225	7x7x25	5,054	19x7x38	12,691	37x7x49
* 300,000	---	735	7x7x15	2,989	7x7x61	7,581	19x7x57
250,000	---	637	7x7x13	2,499	7x7x51	6,384	19x7x48
211,600	0000	532	19x28	2,107	7x7x43	5,320	19x7x40
167,800	000	418	19x22	1,666	7x7x34	4,256	19x7x32
133,100	00	342	19x18	1,323	7x7x27	3,325	19x7x25
105,600	0	266	19x14	1,064	19x56	2,646	7x7x54
83,690	1	210	7x30	836	19x44	2,107	7x7x43
66,360	2	161	7x23	665	19x35	1,666	7x7x34
52,620	3	133	7x19	532	19x28	1,323	7x7x27
41,740	4	105	7x15	420	7x60	1,064	19x56
33,090	5	84	7x12	336	7x48	836	19x44
26,240	6	63	7x9	266	7x38	665	19x35
20,820	7	----	----	210	7x30	532	19x28
13,090	9	----	----	133	7x19	336	7x48
6,530	12	----	----	----	----	168	7x24

1/ The designations of strand construction in accordance with table IV indicated by means of the last number (at the right hand) the number of individual wires contained in the smallest bunch stranded member of the conductor. The preceding number (to the left of the "X") indicates the number of these stranded members which are cabled together to form a rope stranded conductor or a rope stranded member of a larger conductor. A third number preceding these indicates the number of these rope stranded members which are cabled together to form a larger rope stranded conductor. For example, 19x7x25 indicates that 25 wires are bunched stranded into a stranded member; 7 of these members are stranded into a rope lay member and then 19 of these 7 by 25 members are formed into a larger rope stranded conductor having a total of 3325 wires. Strand construction other than specified above shall be approved by NAVSEA."

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Table VIII. Delete and Substitute:

"TABLE VIII. NAVY standard sizes

				Conductor cross sectional area (circular mils)		Maximum conductor resistance D.C. per 1000 ft at 25°C		
Conductor size NAVY Standard	No. of stds (min)	Std dia. nom. (inch)	Cond. dia. nom. (inch)	Nom.1/	Min.	Bare	Coated	Wt. per 1000 ft., lbs appr.
Concentric lay stranded								
* 30 (19)	19	.040	.202	30,860	30,240	.358	.365	95
* 125 (61)	61	.045	.407	124,900	122,400	.0888	.0904	390
* 250 (61)	61	.064	.577	250,500	245,500	.0444	.0453	770
* 350 (91)	91	.062	.682	349,800	342,800	.0316	.0321	1,100
400 (127)	127	.057	.742	413,600	405,400	.0268	.0273	1,300
Bunch lay stranded								
9(90)	90	.010	.120	9,045	8,864	1.22	1.28	28
14(140)	140	.010	.145	14,070	13,790	.786	.823	43
Rope lay stranded								
42(209)	209	.014	.260	42,100	41,280	.272	.284	130
60(304)	304	.014	.310	61,200	60,040	.187	.196	190
83(418)	418	.014	.380	84,230	82,560	.136	.142	270
133(648)	684	.014	.480	137,800	135,100	.0830	.0867	440
150(760)	760	.014	.510	153,100	150,100	.0747	.0780	490
200(998)	998	.014	.580	199,100	195,100	.0575	.0600	630
250(1254)	1254	.014	.680	252,700	247,700	.0453	.0472	800
400(2052)	2052	.014	.850	413,500	405,300	.0277	.0289	1,300
800(4033)	4033	.014	1.150	812,700	796,500	.0141	.0148	2,600

"

1/ Values are for information only.

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3.4.16 Delete and Substitute:

"3.4.16 Cable or surface marking. Cable or cord jacket surface marking is required, unless otherwise specified (see 3.1). Ink marking shall be used. The legend shall be printed in contrasting color: preferably white ink on black or dark background or black ink on white or light background. When indented or embossed marking is specified by the specification sheet, the depth shall not be greater than 0.020 inch. The legend shall consist of the manufacturer's name, the cable or cord type and size designation, when applicable, the part number, year of manufacture and jacket type. The legend shall be repeated at intervals not exceeding 1 foot and the year of manufacture need not be in line with the balance of the legend. For cable constructions on which armor is to be applied, the unarmored or armored cable designations may be applied on the cable jacket. The marking on the cable jacket does not require the use of an "A" for armored or "U" for unarmored cable constructions, for example, both LS3SWU (Unarmored) and LS3SWA (Armored) may be marked as LS3SW.

The character type may be either block or italic and shall have a minimum height in accordance with the diameter over which it is applied as follows:

Diameter range (inch)	Height of character approximate (inch)
0.125 to 0.200	0.047
.200 to .285	.063
.285 to .350	.078
.350 to .500	.094
.500 and larger	.125

Jacket material names shall be abbreviated as follows:

Crosslinked polyolefin	-	XLPOLYO

For example: "Manufacturer's name - LS3SW 19 M24643/36-05 1991 XLPOLYO"

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TABLE XIV Delete and substitute:

"TABLE XIV. Quality conformance inspection. (see 6.2)

Examination or test	Requirement paragraph	Test method paragraph
Basic electrical:		
Conductor resistance	Specification sheet	4.8.4
Voltage withstand	Specification sheet	4.8.8
Insulation resistance	Specification sheet	4.8.5
Conductor and shield continuity	Specification sheet	4.8.10
Jacket flaws	Specification sheet	4.8.9
Group A:		
Visual and dimensional	3.3,3.4,3.5,3.6,3.7 & specification sheet	4.6
Capacitance	Specification sheet	4.8.2
Characteristic impedance	Specification sheet	4.8.3
Watertightness	Specification sheet	4.7.24
Hydrostatic (open end)	Specification sheet	4.7.17
Crack Resistance	Specification sheet	4.7.33
Group B:		
Cold bending cord	Specification sheet	4.7.10
Cold bending cable	Specification sheet	4.7.9
Cross link proof test	Specification sheet	4.7.34
Drip	Specification sheet	4.7.12
Tear strength	Specification sheet	4.7.22
Physicals (unaged)	Specification sheet	4.7.31
* Attenuation	Specification sheet	4.8.1
Gas Flame (1 hour)	Specification sheet	4.7.15
Bending endurance	Specification sheet	4.7.4
Breaking strength	Specification sheet	4.7.5
Twisting endurance	Specification sheet	4.7.23
Group C:		
Physicals (aged)	Specification sheet	4.7.20
Permanence of printing (conductor insulation)	Specification sheet	4.7.18
Permanence of printing (jacket)	Specification sheet	4.7.19
Armor	Specification sheet	4.7.3
Heat distortion	Specification sheet	4.7.16
Cable fill removability	Specification sheet	4.7.8
Shrinkage	Specification sheet	4.7.32
Mutual inductance	Specification sheet	4.8.6
Surface transfer impedance	Specification sheet	4.8.11
Pulse response time	Specification sheet	4.8.7
Tensile & Elongation	Specification sheet	4.7.31
Group D:		
Flame	Specification sheet	4.7.14

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4.7.15.4 Delete and substitute:

"4.7.15.4 Observation. Specimen failure shall be construed if any fuse blows or if any overload relay trips prior to completion of the specified test period (see 3.1)."

4.7.20.3 Delete and substitute:

"4.7.20.3 Procedure.

(a) Air oven accelerated aging: each specimen shall be freely suspended vertically, secured by one end, within the oven. The air temperature within the oven shall then be raised to the value specified in table XVIII, and shall be maintained at this value for a continuous period of not less than that specified in table XVIII. Each specimen shall then be removed from the oven and allowed to return to room temperature and then tested as specified in 4.7.31."

TABLE XVIII Delete and substitute:

"TABLE XVIII. Details for accelerated aging procedures.

Specimen material	Procedure	Temperature (°C)	Minimum period (Hours)
Insulation Specimens:			
Polyethylene (cross linked)	Air oven	121 ± 2	168
Rubber (Ethylene Propylene)	Air oven	121 ± 2	168
Jacketing Specimens:			
* Silicon rubber	Air oven	260 ± 2	24
* Cross linked	Air oven	136 ± 2	168
* Polyolefin	Hot oil immersion	121 ± 2	18

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4.7.27.2 Delete and substitute:

"4.7.27.2 Special procedures. Special procedures shall include the following:

A. Only the use of propane gas shall be allowed.
B. The burner setting shall be : Air 500^{cc}/min., Gas 50^{cc}/min.
C. The Radiometer shall be used to calibrate the heat flux at the initial start of each test period.
D. The test chamber shall be calibrated using a NIST CELLULOSE standard at the initial start of each test period.
E. The chamber shall be inspected to assure airtightness. Seals and gaskets shall be replaced as needed.
F. The test shall be run using a minimum of three specimens. If test values are below the limits specified (see 3.1) test is complete. If one test value deviates from the median value by more than 20 percent, two additional tests shall be performed. The high and low test values shall be discarded and the remaining three averaged."

TABLE XX Delete and substitute:

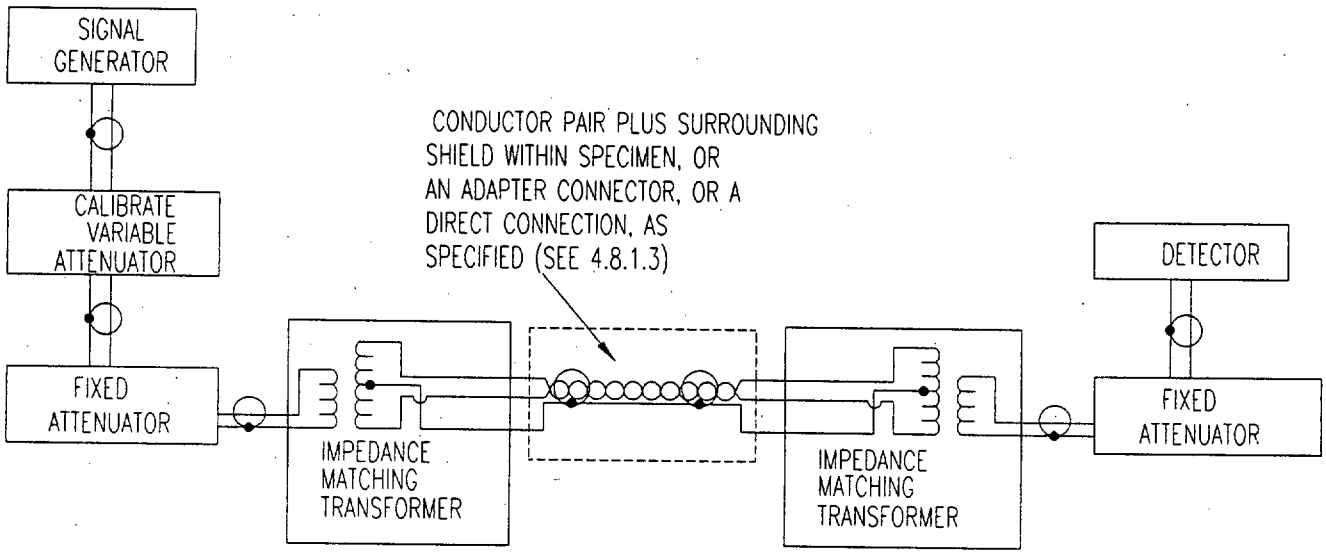
"TABLE XX Immersion Test temperatures.

	Fluid	Test temperature
	Fuel oil, MIL-F-16884	98 to 100°C
	Turbine fuel, JP-5, MIL-T-5624	48 to 50°C
*	Cleaner, Isopropyl alcohol, TT-I-735	20 to 25°C
	Hydraulic fluid, MIL-H-5606	48 to 50°C
	Hydraulic fluid, MIL-H-17672	48 to 50°C
	Lubricating oil, MIL-L-23699	98 to 100°C
	Coolant, Monsanto Coolanol 25 or equivalent	20 to 25°C

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FIGURE 17 Delete and substitute:

"



SH 13066

FIGURE 17. Block diagram of electrical connections for the attenuation test when testing a conductor pair plus surrounding shield."

Custodians:
Army - MI
Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Preparing Activity:
Navy - SH

Agent:
DLA-IS

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